



Unlocking the World of Probability

Section 1: Learn

What is Probability?

Probability is the **measure of how likely an event is to happen**.

It helps us **quantify uncertainty** and make **predictions based on data**.

Why is Probability Important?

- **Used in Daily Life** – Weather forecasts, elections, and games of chance.
- **Supports Decision-Making** – Businesses use probability for **risk analysis**.
- **Foundation for Statistics & Machine Learning** – Helps in **predictive modeling**.

Basic Probability Formula

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$$P(\text{Event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

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- **Always between 0 and 1.**
- **0** means the event **will not happen**.
- **1** means the event **will definitely happen**.

Types of Probability

1. **Theoretical Probability** – Based on **mathematical calculations**.
 - *Example:* Probability of rolling a **6 on a die** = **1/6**.
2. **Experimental Probability** – Based on **actual experiments**.



- *Example:* Tossing a coin 100 times and noting how many times heads appear.

3. **Conditional Probability** – Probability of an event **given another event has occurred**.

- *Example:* Probability of picking a red ball **after removing a blue ball**.

How is Probability Used?

- **Insurance Companies** – Estimate accident risks.
- **Stock Market** – Predict price movements.
- **Cricket Analysis** – Probability of winning based on past performances.

Anecdote: How Probability Helped in the Cricket World Cup

During a T20 World Cup, analysts **calculated the probability of a team winning** based on past match statistics.

By analyzing **team performance, pitch conditions, and weather**, they provided **accurate predictions**, helping teams plan their strategies better.

Section 2: Practice

1. Calculating Basic Probability

Example: Tossing a Fair Coin

Outcome	Probability
Heads	1/2



Outco me	Probabil ity
Tails	1/2

Observation: Each side has an **equal chance of 50%**.

2. Probability of Rolling a Die

Die Face	Probabil ity
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

- **P(rolling an even number) = $3/6 = 1/2$** (since even numbers are 2, 4, 6).
 - **P(rolling a number greater than 4) = $2/6 = 1/3$** (since 5 and 6 are greater than 4).
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3. Conditional Probability Example

Example: Drawing a Red Ball from a Bag

Col or	Cou nt
Red	4



Col or	Cou nt
Blue	6
Total	10

- $P(\text{Drawing a Red Ball}) = 4/10 = 0.4$ (40%)
 - If one red ball is removed, $P(\text{Drawing Another Red}) = 3/9 = 1/3$ (33.3%).
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Section 3: Know More

Frequently Asked Questions (FAQs)

1. What is the difference between Theoretical and Experimental Probability?

- **Theoretical** is based on logic (e.g., rolling a die).
- **Experimental** is based on actual trials (e.g., recording real outcomes).

2. Can probability be greater than 1?

No, probability always ranges between 0 and 1.

3. What is the probability of an impossible event?

The probability of an impossible event is 0.

4. What is the probability of a certain event?

A certain event has a probability of 1.



5. How does probability relate to real-life decision-making?

It helps in **risk analysis, business forecasting, and predicting uncertain events.**

Conclusion:

Probability is a **fundamental concept** used in **decision-making, science, and machine learning.**

By understanding **basic probability, conditional probability, and real-world applications,**

we can **analyze uncertain situations and make better predictions.**